

iDecide: Leveraging Deep Learning to Enhance Seamless Handover Decisions in B5G Networks

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Abstract—Ensuring seamless connectivity in B5G networks critically depends on making accurate and proactive handover decisions. However, existing learning-based methods often fail in SDN-enabled dynamic environments due to limited interoperability and insufficient evaluation frameworks. This letter addresses this gap by introducing iDecide, a DL-based data-driven framework to optimize handover decisions in B5G networks with SDN support. The proposed hybrid architecture uses a dual-path deep learning framework, incorporating residual connections and feature engineering to improve learning efficiency and decision accuracy. Achieving up to 99.06% validation accuracy, iDecide not only identifies the optimal handover target cell with high precision but also significantly reduces end-to-end delay by 33% and achieves a peak bandwidth of 16.70 Mbps, as demonstrated in an SDN-enabled testbed.

Index Terms—Decision-making, Seamless handover, DL, QoS, SDN, B5G.

I. INTRODUCTION

WITH the emergence of Beyond 5G (B5G) technology, the demand for high-quality services proliferates, especially in urban areas that have several B5G applications such as banking, transportation, healthcare, critical services, etc [1]. These applications require continuous internet delivery, where Seamless Mobility Management (SMM) is crucial. SMM proactively manages user mobility and network resources efficiently through decision making to improve Quality of Service (QoS) [2]. Furthermore, software-defined networks (SDN) and learning models pose as key enablers to improve QoS in future networks [3]. Adopting SDN allows for the agile integration of learning models and provides a global view of the network with centralized control. An SDN-enabled B5G infrastructure consists of several devices (such as mobile phones, tablets, smart devices, etc.) termed User Equipment (UE), open flow switches, a central controller, heterogeneous base stations called Next Generation Node B (gNB), multiple communication links, and various protocols. These UEs are connected to gNBs through an SDN-enabled heterogeneous cellular network to access the internet. In addition, any changes in this network due to mobility, signal quality, and load distribution encounter service outages due to frequent handovers. This leads to repeated decision making and updates to flow rules, showing general delays and poor QoS [4]. Therefore, achieving SMM in SDN is challenging and raises the essential need to integrate learning-based handover mechanisms [5].

Motivation: In the state-of-the-art, researchers have proposed several approaches to optimize the handover process and improve QoS. Among these, learning model-based solutions, such as Machine Learning (ML), Deep Learning (DL), and Reinforcement Learning (RL), are very prominent [6]. Most of the work focuses on the incorporation of ML and DL for handover decisions, considering only signal parameters [7], [8]. However, enhancing decision-making requires a global view of the network to efficiently manage the overall resources. Without this, the approaches become inaccurate and lack precise performance evaluation within an SDN environment. Furthermore, some solutions that integrate an RL-based approach fail to handle complex data and dynamic mobility scenarios such as high-speed trains, etc., leading to suboptimal decisions and increased outages [9], [10]. Lately, work [11] introduces LISP-MM, which improves overall QoS in SDN, assuming that wireless links are reliable, which might not hold in all SDN-based B5G deployments, particularly in heterogeneous networks. Whereas, we have proposed a DL-based dual-path framework for combining both capabilities to handle handover trigger and decision-making in a single module. Its modular architecture makes it compliant with SDN-enabled networks and real-time decision-making in dynamic environments, which has not been addressed in prior works.

Through this letter, we have made focused efforts to address and solve the following primary research questions:

- Q1) How can we achieve seamless mobility leveraging DL-based mechanisms in SDN-based B5G cellular networks?
- Q2) What is the performance impact of the DL-based solution on the QoS, considering the dynamic mobility of UEs?

In summary, we have achieved seamless handovers in B5G with this letter, making the following contributions:

- 1) We have introduced a novel dual-path DL-based framework for handover decisions that considers both signal and overall resource load metrics in SDN-enabled B5G networks.
- 2) Our approach integrates proactive handover triggering with a data-driven algorithm to select an optimal gNB cell, specifically tailored for dynamic mobility scenarios.
- 3) As proof of concept, we have deployed the solution in an SDN-enabled testbed for a precise evaluation in B5G, achieving an impressive validation accuracy up to 99.06%.

The rest of this letter is organized as follows: Section (II) presents a mathematical problem formulation. Section (III) details the proposed solution. Section (IV) discusses the details of testbed implementation, followed by the result analysis. Lastly, Section (V) concludes work and outlines future scope.

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II. PROBLEM FORMULATION

Let us assume an SDN-Enabled B5G network having m number of gNBs, each serving a set of UEs $u = \{u_1, u_2, \dots, u_n\}$ where n is the total no. of operating UEs. This network operates on a cellular architecture that monitors real-time signals and performance metrics. For each UE $u_i \in u$ the recorded UE metrics at time t as expressed in Eq. (1) by

$$\mathbf{M}_i(t) = [\text{RSRP}_i(t), \text{RSRQ}_i(t), \text{PRB}_i(t), D_i(t)]^\top \quad (1)$$

where RSRP is the Reference Signal Received Power, RSRQ is the Reference Signal Received Quality, PRB is Physical Resource Blocks, and D is the physical distance between the UE and gNBs. This follows a multivariate Gaussian distribution ($\mathcal{N}$) represented by $\mathbf{M}_i(t) \sim \mathcal{N}(\boldsymbol{\mu}_i, \Sigma_i)$ where $\boldsymbol{\mu}_i$ is the mean vector and Σ_i is the covariance matrix.

In normal operating conditions, the signal measurement metrics are recorded from the network by substituting the values from Eq. (1) into Eq. (2), which can be defined as

$$\mathbf{M}_{\text{normal}}(t) = \mathbf{M}_i(t) + \epsilon_i(t) \quad (2)$$

where $\epsilon_i(t) \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ represents Gaussian noise, where $\mathbf{0}$ is mean vector and $\sigma^2 \mathbf{I}$ is covariance matrix.

Beyond normal operating conditions, various outage scenarios impacting seamless mobility, derived from Eq. (2) are:

- 1) **Signal degradation:** It occurs when RSRP exhibits a time-dependent decay due to interference or fading. This can be modeled using an exponential decay function as

$$\mathbf{M}_{\text{degraded}}(t) = \mathbf{M}_{\text{normal}}(t) - \alpha e^{-\lambda t} \quad (3)$$

where α represents the initial degradation amplitude, and $\lambda > 0$ is the decay rate.

- 2) **Signal aberrations due to mobility:** High mobility introduces stochastic variations in RSRP and RSRQ given by

$$\mathbf{M}_{\text{mobility}}(t) = \mathbf{M}_{\text{normal}}(t) + \mathbf{F}(t)\boldsymbol{\xi} \quad (4)$$

where $\mathbf{F}(t) \sim \mathcal{N}(\mathbf{0}, \delta^2 \mathbf{I})$ is a random perturbation matrix with variance δ^2 , and $\boldsymbol{\xi}$ is a unit-norm vector.

- 3) **Signal blackout:** In severe disruptions, metrics freeze at constant values due to a lack of coverage, modeled as

$$\mathbf{M}_{\text{blackout}}(t) = \mathbf{1} - c \quad (5)$$

where $\mathbf{1}$ is a unit vector, and c is a constant scalar.

- 4) **Resource load imbalance:** In a load imbalance scenario, PRB usage $\text{PRB}_i(t)$ increases due to the uneven distribution of UEs between cells varies spatially as

$$\mathbf{M}_{\text{overloaded}}(t) = \mathbf{M}_{\text{normal}}(t) + \beta \cdot \phi(d_i) \quad (6)$$

where β is the overload factor, D_i is the distance of u_i from the serving gNB, and $\phi(D_i)$ is a spatial function given by

$$\phi(d_i) = \frac{1}{1 + \exp(-\gamma(D_i - D_0))} \quad (7)$$

where γ controls the sharpness of the distance-dependent load effect, and D_0 is a threshold distance.

III. PROPOSED SOLUTION

Our proposed solution focuses primarily on enhancing the seamless and ubiquitous nature of the cellular network by improving QoS. A comprehensive workflow is represented in Fig. 1 along with the step-by-step (Steps 1-8) execution of the proposed approach. For this, we have used a dataset which was

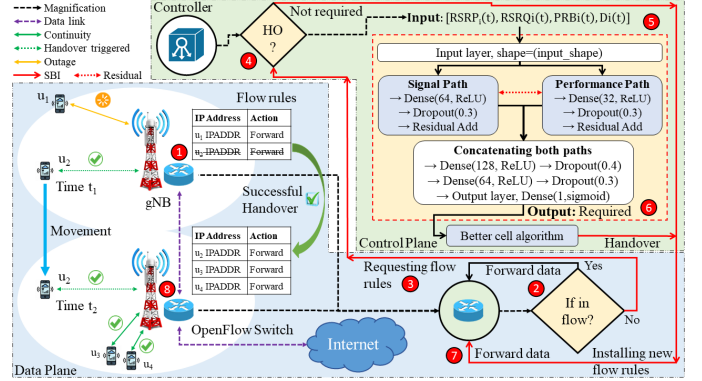


Fig. 1: Illustration of workflow

generated using MininetWiFi [12], containing 10,000 entries measured every 10ms. A scenario of six neighboring cells and one serving cell, where the UEs have a random mobility model, was considered. The dataset label ue-id determines the distinct mobility conditions, such as waiting passenger, pedestrian, car, train passenger, etc. In addition, it contains 8000 entries where QoS is satisfactory, and 2000 have poor QoS. A poor QoS condition can be represented by

$$\text{QoS}_{\text{poor}}(t) = \text{QoS}_{\text{normal}}(t) - \Delta \text{QoS}(t) \quad (8)$$

where the combined lowering rate for all outage scenarios, substituting the values from Eq. (3), (4), (5), and (6) can be expressed as $\Delta \text{QoS}(t) = \alpha e^{-\lambda t} - \beta \varphi(D_i) + (1 - c) + F(t)\xi$. Thus, for an optimal solution, $\Delta \text{QoS}(t)$ should be minimum.

A detailed overview of the proposed solution, offering comprehensive insights, is given in the following subsections:

A. Methodology

For a given neighboring cell $k \in m$, where m = set of candidate gNBs, represented by metric vector $\mathbf{M}_k$, with the objective to select a better cell for higher QoS. The decision is based on a scoring function by integrating Eq. (1) and (8)

$$\text{score}_k = w \cdot \mathbf{M}_k + \int_{t_1}^{t_2} \text{QoS}_k(t) dt \quad (9)$$

where $w = [w_{\text{RSRP}}, w_{\text{RSRQ}}, w_{\text{PRB}}, w_D]^\top$ are weight coefficients, $\text{QoS}_k(t)$ is the QoS function over the time interval $[t_1, t_2]$. The best cell is selected by utilizing Eq. (9) as

$$\text{cell}_{\text{best}} = \arg \max_k \{ \text{score}_k \mid \mathbf{M}_k \text{ satisfies constraints} \} \quad (10)$$

having values for $\mathbf{M}_k$ from Eq. (1), satisfying $\text{RSRP}_k > \theta_{\text{RSRP}}$, $\text{PRB}_k < \theta_{\text{PRB}}$, and $D_k < \theta_D$, where θ_{RSRP} , θ_{PRB} , and θ_D as defined thresholds. The effective way to select the best cell is to optimize the core network management, which leverages an SDN switch that follows flow rules to determine

routing actions, i.e., forward or drop. To optimally install these flow rules, we introduce iDecide, a novel DL-based dual-path decision-making framework to trigger the handover process and find the best target cell based on Eq. (10). Initially, the SDN switch analyzes the packets in the data plane and checks whether the packets are in a flow. If the packets are not in flow, then the control plane is requested to install the requisite flow rules through the south-bound interface. Here, iDecide analyzes the location, signal measurements, and PRB usage of serving along with neighboring cells to check whether a handover is required. If not required, then the flow rules are updated for forwarding packets that are not in the flow. Otherwise, a soft handover (make-before-break) is initiated, and the neighboring cells are evaluated to find an optimal one. As soon as a better cell is found, the flow rules are installed, and communication happens. However, if no better cells are found, then the existing link is retained. This framework efficiently distributes the traffic across the access network and thus improves the overall QoS in dynamic environments.

B. Dual-path decision framework

Our novel framework, incorporated in the control plane, has a central view of the whole network, which helps to efficiently extract the signal measurements and resource usage features. Next, these features are fed to the decision-making model. Here, the integration of a dual-path processing architecture is introduced, separating the feature analysis into distinct paths, one for complex signal features processing, which is dedicated to analyzing multiple signal measurements in dynamic environments, and the other for optimizing the analysis of network resource metrics. Further, the residual connections between paths prevent vanishing gradient issues and retain the importance of features, which ensures improved training without performance degradation. The dual paths process data in parallel dense layers, with output concatenated for final analysis, with fast convergence within 15 epochs, and achieve impressive validation accuracy up to 99.06%. This combines both handover triggering and finding better cells into a single model. Furthermore, its efficient layered design integrates a handover decision-making algorithm, enabling robust real-time handover decisions with significant operational advantages.

C. Optimal handover decision-making algorithm

We have developed a robust algorithm adapted for dynamic environments, as illustrated in Algorithm 1. The mechanism responds to service outages by leveraging a multi-metric optimization approach to manage mobility and ensure service continuity. It begins with computing the change in QoS, $\Delta QoS_i(t)$ (line 2), which compares the current performance of the current cell ($cell_i$) against predefined thresholds θ_t . If the values are not satisfactory, our algorithm evaluates neighboring cells by calculating a multi-metric score (lines 4–5). These scores incorporate normalized and weighted metrics, including signal strength (RSRP, RSRQ), cell load (PRB usage), and distance between the UE and each gNB. Candidates below the QoS threshold are discarded (line 6), and the cell having the highest score is selected as the best target cell (line

7). If valid, the handover is executed (line 9); otherwise, the connection is retained (line 11). This adaptive approach minimizes interruptions and improves overall QoS.

Algorithm 1 Cell selection and handover decision-making

Require: $M_i(t)$, $\{M_k(t)\}$, $w_i(t)$, $\theta_i(t)$
Ensure: $cell_{best}$

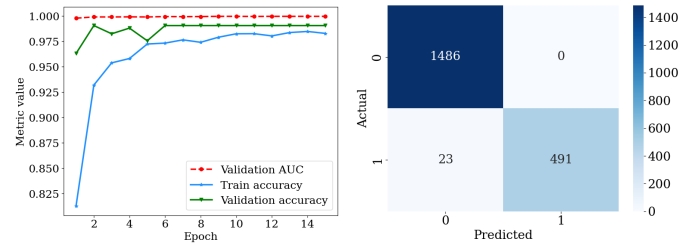
- 1: Calculate $\Delta QoS_i(t)$
- 2: **if** $\Delta QoS_i(t) > \theta_t$ **then**
- 3: **for** each neighboring $cell_k$ **do**
- 4: Compute $score_k \leftarrow M_k(t) \in \{M_k(t)\}$
- 5: **end for**
- 6: Filter $\{cell_k\} \ni \Delta QoS_k(t) > \theta_t$
- 7: Select $cell_{best} \in \{cell_k\}$ having $\max\{score_k\}$
- 8: **if** $cell_{best}$ is valid **then**
- 9: Handover to $cell_{best}$
- 10: **end if**
- 11: **else**
- 12: Retain $cell_i$
- 13: **end if**

IV. TESTBED IMPLEMENTATION AND RESULT ANALYSIS

As a proof of concept in an SDN-enabled B5G network environment, we employed *Mininet-WiFi* [12], deployed on a dedicated system equipped with an Intel Core i7-10700 2.90GHz CPU, 16GB RAM, an AMD Radeon RX 640 discrete GPU, and a 1 TB SSD running Ubuntu 20.04 LTS. To mimic the physical characteristics of wireless media, we integrated *wmediumd* [13], which supports advanced interference modeling with configurable parameters. In this setup, the access point (AP) simulates a gNB, while the station (STA) represents a UE, communicating over the wireless medium. Furthermore, network traffic is generated using the *iPerf3* tool [14], allowing simulation of realistic B5G scenarios in SDN. A summary of the configured parameters is given in Table I.

TABLE I: Configured parameters for performance evaluation

Parameter	Value
gNBs (count/range/Tx power)	13 / 100, 500, 1000 m / 31, 62, 76 dBm
Controllers / Server	1 (Central OVSController) / 2 (Internet)
Propagation model	Log-distance, Exp. 4.5
Mobility model	Random direction
UEs (count/speed/Tx power)	20 / variable / 14 dBm
UE protocol	TCP, MPTCP



(a) Accuracy & AUC over epochs

(b) Confusion matrix

Fig. 2: An illustration depicting model performance

Based on the performance metrics captured, visualizations obtained to validate our solution are shown in Fig. 2. First, we

have evaluated the decision-making framework through the DL model performance metrics. The graph in Fig. 2a illustrates the training and validation accuracy along with AUC over 15 epochs. Both accuracy and AUC show fast convergence, with high validation metrics indicating strong generalization. In addition, it achieves an optimal AUC and a validation accuracy of 99.06%. Furthermore, the confusion matrix given by Fig. 2b indicates no false positives, underscoring the robustness of the model. This proves its effectiveness in minimizing false alarms and ensuring a proactive handover decision-making.

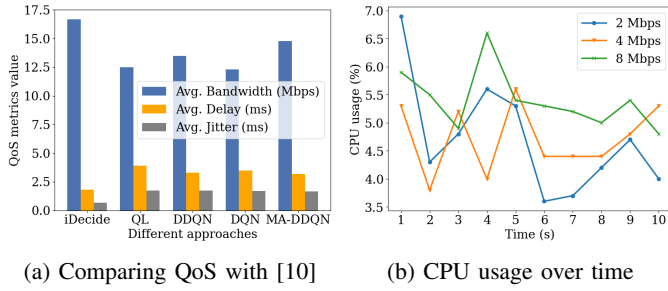


Fig. 3: Comparing performance with different approaches

Then, in Fig. 3 we have compared the QoS performance metrics of iDecide with state-of-the-art solutions. Here, Fig. 3a illustrates the variance of QoS metrics with work [10]. In addition, the experimental results demonstrate that our solution outperforms, with the lowest average delay of 1.81 ms and jitter of 0.68 ms, which are nearly optimal. Furthermore, our solution achieves a peak bandwidth of 16.70 Mbps. Also, the CPU usage trends in Fig. 3b exhibit a low computational overhead for multiple bandwidths. Therefore, we can conclude that our solution is suitable and ensures seamless connectivity in dynamic environments while enhancing overall QoS.

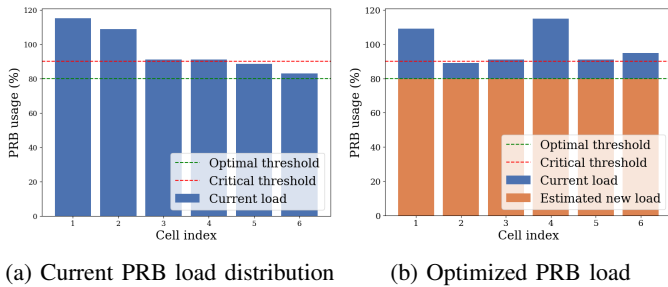


Fig. 4: Analyzing the impact of our solution on resources

Lastly, we analyzed PRB usage, a key factor contributing to load imbalance, which leads to outages. The distribution is illustrated in Fig. 4. The instantaneous distribution of PRB load, where Fig. 4a suggests that the majority of candidate cells operate above the optimal threshold and very close to critical. This implies a resource imbalance in the overall network. However, Fig. 4b represents the significant reduction in PRB usage by integrating our solution. This provides visual validation that iDecide not only enhances QoS but also efficiently balances the PRB load over heterogeneous cells. Thus, we can infer that iDecide ensures optimal performance and avoids reaching the critical PRB threshold.

V. CONCLUSION AND FUTURE SCOPE

We have improved the handover decisions for seamless connectivity in SDN-enabled B5G networks, using DL, resulting in a high validation accuracy of up to 99.06%. Our experimental results demonstrate the ability to achieve optimality in the management of dynamic mobility scenarios. Impressively, iDecide reduces the average delay by 33% and achieves a maximum bandwidth of 16.70 Mbps. Future work focuses on integrating distributed controller capabilities into this work.

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